THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 10

Dynamics and Approximation: The Structure of Change

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. The two ingredients of a dynamical system

a. State, [what is the minimal state of a system from your own work?]
b. Rule of evolution:

____

____

2. "The rule f is a __________ on the state space."

A trajectory is what you get by:

____

____

3. The three kinds of fixed point

Stable:
Unstable:
Saddle:
Stability is a property of ____________, not of the equilibrium value.

4. Linearization

Near a fixed point, Taylor's theorem gives $d\delta/dt \approx$ ____________
The eigenvalues of the Jacobian tell you:
negative real part $\rightarrow$
positive real part $\rightarrow$
imaginary part $\rightarrow$

5. Why linear methods are unreasonably effective (in one sentence,

using the word "smooth"):

6. The four genuinely nonlinear phenomena, each with an example

from biology or my own work:
Multistability:
Oscillation (limit cycle):
Bifurcation:
Chaos:

7. Feedback: "feedback loop" = "the sign of the ________."

Negative feedback builds:
Positive feedback builds:

8. A wall I have hit (or expect to hit) in my own work that is

really a nonlinear-dynamics wall: